Phantom homes: one in three homes destroyed by recent wildfires is still on the map

Best Bias Discovery entry · Zindi user Arnaud_TOE · Bias Bounty Mapping Equity Challenge *Version 1.0, 2026-10-23. Associated leaderboard submission: `iu1dZ8Hf` (public RMSE 0.078204). Code and outputs: https://github.com/Arnaud-TOE/bias-bounty-phantom-homes. Two commands regenerate every number (Section 7).*

The finding in one paragraph

In the Northern California study region, CAL FIRE inspected every structure inside the perimeters of recent wildfires. Of the **6,868 structures destroyed by fires from 2020 to 2025**, **2,063 (30.0 %) are still drawn as standing buildings** in Overture release 2026-08-19.0. Each of these footprints has a source date (Microsoft imagery date or OpenStreetMap last edit) **earlier than the day the fire started**, so none of them can be a rebuilt home. Among them are **1,421 dwelling units**. Net of chance proximity (a placebo 60 m away), the excess is **24.3 % (95 % CI 17.4–35.8 %, clustered by fire), about 1,670 structures**. For fires up to 2019 the same rate is **0.2 %**: the map eventually catches up, but for five years after a disaster it keeps showing neighbourhoods that no longer exist. Greenville (Dixie Fire, 2021), Berry Creek (North Complex, 2020), Cohasset (Park Fire, 2024) and the Karuk communities of Happy Camp and the Klamath River carry hundreds of these *phantom homes*.

1. Why the automated scorecard cannot see it

- **The coverage gap is one-sided.** Every component is `1 - min(1, Overture / reference)`. A map that shows *more* than exists is scored as perfect coverage, so phantom structures can only hide gaps, never reveal them.
- **Burn scars look well mapped.** A tract whose homes burned but stay drawn keeps its full building count. The scorecard's wildfire stratum therefore reports these tracts as covered. If the building reference also predates the fire, the building component simply compares two pre-fire maps.
- **No stratum measures time since disaster.** The five metrics stratify by urban/rural, SVI, CVI, tribal status and hazard exposure. None of them asks whether the map was updated after the hazard actually struck.

2. Evidence

Data.

- CAL FIRE Damage Inspection (DINS) points: one record per inspected structure, with damage class, incident name and start date (49,168 points inside the northern-ca scored tracts).
- Overture buildings from the challenge extract.
- Tract strata from the challenge package.
- Sources and checksums are listed in Section 7.

Definition (deliberately conservative). A destroyed structure is a *phantom* when:

1. an Overture footprint lies within 10 m (exact distance in California Albers, EPSG:3310);
2. that footprint's source date is known (the Microsoft placeholder date 2000-01-01 is excluded);
3. that date is earlier than the fire's start date.

Footprints dated after the fire might be rebuilt homes and are never counted. In downtown Greenville this excludes 257 destroyed structures that still sit under a footprint (Figure 2, orange), so the true number of phantoms is higher than reported.

Main result

	Fires ≤ 2019	Fires 2020–2025
Destroyed structures (DINS)	25,594	**6,868**
Phantoms within 10 m	59 (0.2 %)	**2,063 (30.0 %)**
…of which dwelling units	40	**1,421**
Point inside the footprint (0 m) / 5 m / 20 m	0.2 % / 0.2 % / 0.3 %	22.1 % / 27.4 % / 34.0 %
Placebo: same points shifted 60 m (seeded)	0.04 %	5.8 %
Excess over placebo (95 % CI, bootstrap over 85 fires)	—	**24.3 % (17.4–35.8 %), ≈ 1,670**
Control: standing structures matched within 10 m	85.3 %	69.2 %
Source of the phantom footprint: Microsoft ML / OpenStreetMap	—	1,602 / 461

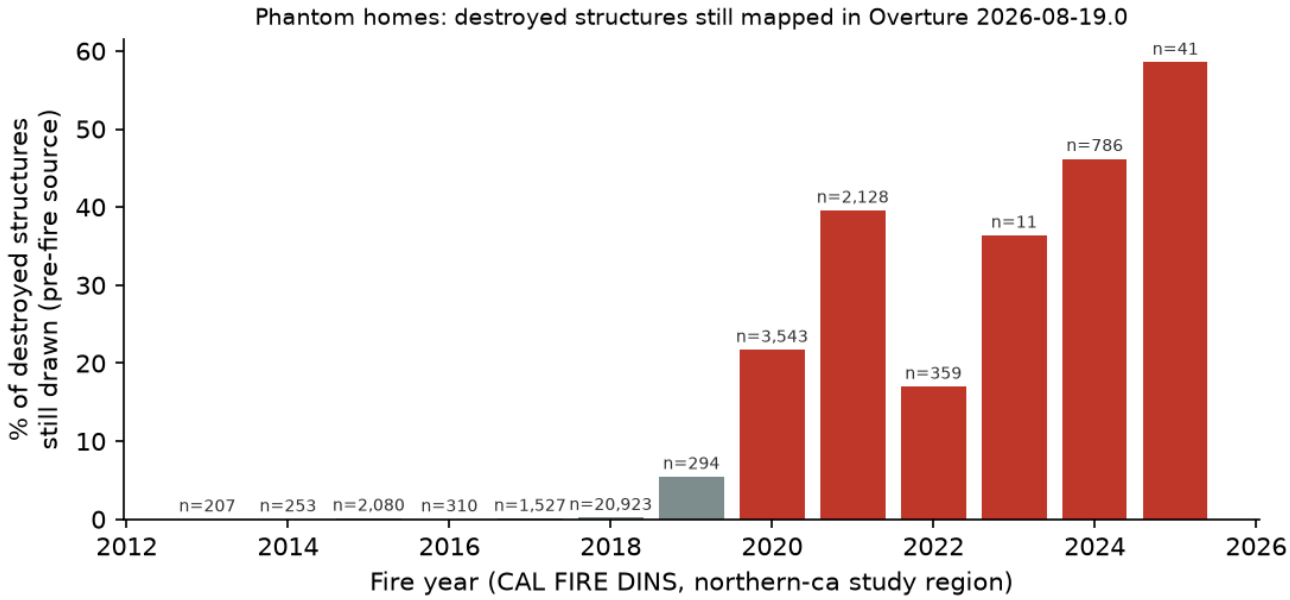


Figure 1. Share of destroyed structures still drawn with a pre-fire source, by fire year (red: 2020 and later).

By fire (destroyed / phantoms / share):

- Dixie 2021: 1,311 / 471 / 36 %
- North Complex 2020: 2,352 / 398 / 17 %
- Park 2024: 713 / 323 / 45 %
- LNU Lightning Complex 2020: 364 / 264 / 73 %
- Fawn 2021: 185 / 127 / 69 %
- Slater 2020: 353 / 55 / 16 %
- Cache 2021: 60 / 53 / 88 %
- Camp 2018 (Paradise): 18,804 / 29 / **0.2 %**. The largest disaster *was* cleaned from the map; the smaller, more remote ones were not.

Named communities (phantoms, dwelling units):

- Greenville, Plumas Co. (350, 233)
- Berry Creek, Butte Co. (165, 110)
- Cohasset, Butte Co. (153, 109)
- Napa (128, 111)
- Redding (126, 66)
- Happy Camp, Siskiyou Co. (55, 40)
- Chester (39, 28)
- Doyle (37, 26)

Named tracts (`discovery/out/phantom_by_tract.csv`):

GEOID	Area	Phantoms	SVI	Note
06007002402	Berry Creek	382	0.81	Enterprise Rancheria
06063000400	Greenville	357	—	Greenville Rancheria
06055201800	Napa County	262	0.77	—
06007001602	Cohasset	226	—	—
06093001300	Klamath River / Happy Camp	79	0.83	Karuk Off-Reservation Trust Land
06033000702	Clearlake	53	0.97	88 % of destroyed homes still drawn
06035040600	Lassen County	41	0.87	Susanville Off-Reservation Trust Land

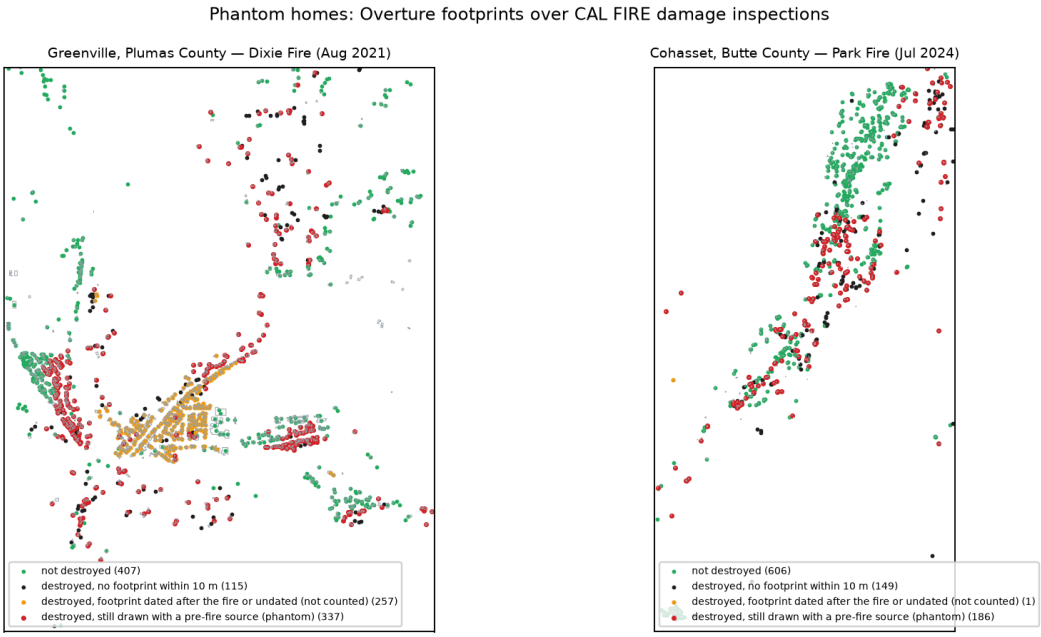


Figure 2. Greenville (Dixie 2021) and Cohasset (Park 2024). Red: destroyed, still drawn with a pre-fire source. Orange: destroyed, footprint dated after the fire (not counted). Black: destroyed, no footprint. Green: not destroyed.

Who is exposed. Recent fires hit the most vulnerable tracts. Of the destroyed structures, 52.5 % stand in tracts in the top SVI quartile band (SVI ≥ 0.75), against 31.3 % of the region's tracts; phantoms follow them (47.6 %, 1.5 times the regional share). Within the same fire (linear probability model with fire fixed effects and tract-clustered errors), the phantom rate does not rise further with SVI ($p = 0.71$). It is 11.5 points higher in tracts that touch a tribal statistical or legal area ($p < 0.001$), although this rests on 17 tracts (`out/phantom_exposure.csv`).

3. Who is affected and what it changes

The people affected are those already displaced: households in Greenville, Berry Creek, Cohasset, Happy Camp and Doyle, many of them in high-SVI tracts and in Rancheria and Karuk trust-land areas. They are still rebuilding, and the next fire season arrives on a map that describes the town as it was before the last one.

- **Evacuation.** Evacuation zones are sized and staffed from structure and population counts. In Greenville the map still carries 350 structures (233 dwelling units) that burned in 2021, so door-to-door notification lists and estimates of people at risk built from it are inflated in exactly the zones that will be ordered out again.
- **Structure triage and dispatch.** Pre-incident plans and structure-protection triage can be built on footprint inventories. An engine assigned to a "structure" that is an empty lot is an engine not protecting a home that was rebuilt next door.
- **Relief and recovery.** Damage-assessment baselines and recovery dashboards that start from the map will count these 1,421 dwelling units twice: as destroyed in 2021, and as standing in the next disaster's baseline. The recovery gap, meaning the people still displaced, becomes invisible.

Why the map stays frozen. Two mechanisms meet in Overture:

- OpenStreetMap practice discourages deleting burned buildings. Mappers are told to re-tag them with a lifecycle prefix (`destroyed:building`), often only once imagery is updated; until someone does, the polygon stays a building. This is the case for 461 of our 2,063 phantoms: OpenStreetMap source, last edited before the fire.
- Microsoft ML footprints come from imagery captured before the fire: 1,602 phantoms, with known imagery dates earlier than the fire's start.

The same frozen footprints feed exposure products. In the Greenville tract (06063000400) the 357 phantoms are **16.5 % of all Overture buildings**. The challenge's own exposure layer, USFS Wildfire Risk to Communities (2nd edition, "2022 conditions", strata column `usfs_BuildingCount_sum`), counts 2,440 buildings there, more than Overture's 2,170 including the phantoms. This is consistent with an exposure layer also built on pre-fire footprints. It is a tract-level comparison, not a proof, and the column is used here only for this check, never in our scored pipeline. The remediation follows directly: re-tag footprints from CAL FIRE damage inspections after each fire, and date ML footprints by imagery so that consumers can mask pre-fire geometry inside new fire perimeters.

4. Robustness

- The result does not depend on the search radius (22–34 % between 0 and 20 m).
- It is not chance proximity: the placebo stays at 5.8 %, and the excess over it is 24.3 % with a 95 % interval of 17.4–35.8 % when whole fires are resampled.
- It is not a matching artefact: standing structures are matched at 69–85 %.
- Rebuilt homes are excluded by construction (source date before the fire).
- The ≤ 2019 contrast shows the method finds almost no phantoms where the map has had time to update.

5. Limits

- One region and one ground-truth source (CAL FIRE covers California only).
- DINS points are inspector-placed and can sit a few metres from the structure; the radius sensitivity bounds this.
- Microsoft source dates are imagery dates assigned per tile.
- Footprints dated after a fire may still be stale, so the count is a lower bound.
- The Eastern Washington fires of 2020 and 2023 are not covered (no equivalent public structure-level inspection data).

6. How this entry relates to other entries

Another entry (#35474) also matches CAL FIRE inspections to Overture footprints. It reports about 1,200 likely stale footprints in zones where the AI imagery predates the fire, and a destroyed-versus-undamaged match ratio of 0.89 (95 % CI 0.72–1.06). We read that as independent confirmation and add five things it does not establish:

1. **The time dimension.** The same test on 25,594 structures destroyed up to 2019 finds 0.2 %. The problem is specific to recent disasters, and the largest one (Camp 2018) was cleaned while smaller, more remote fires were not.
2. **A control-adjusted estimate.** The placebo gives the chance rate directly (5.8 %), hence an excess with a fire-clustered interval that excludes zero (17.4–35.8 %), where a match ratio whose interval includes 1 cannot separate staleness from matching rate.
3. **A per-footprint date rule** (each footprint's own source date against the fire's start), rather than a zone's modal imagery date. It also catches the **461 OpenStreetMap phantoms** (22 %), a separate cause (the lifecycle-tag practice) that imagery dating cannot reveal.
4. **Who is exposed**: 1,421 dwelling units, the SVI and tribal-area contrasts, the named communities, and the 24 tracts with at least 10 phantoms (`out/phantom_by_tract.csv`).
5. **The link to exposure products** and a concrete remediation.

7. Reproducibility

```
python -I get_data.py             # challenge extract (~470 MB) and external files, SHA-256 checked
python -I discovery/reproduce.py  # rebuilds every number, table and figure (~2 min)
```

Source	URL	Retrieved	SHA-256
CAL FIRE DINS (CC-BY)	https://gis.data.cnra.ca.gov/api/download/v1/items/994d3dc4569640caadbbc3198d5a3da1/csv?layers=0	2026-10-17	`af01e965…38404f`
CMS Nursing Home Provider Information (public, annex)	https://data.cms.gov/provider-data/dataset/4pq5-n9py	2026-10-17	`378d76a4…d1e2b0`
Overture 2026-08-19.0 buildings, challenge extract	`reference/northern-ca/northern-ca-overture-buildings.parquet`	2026-10-16	SHA-256 in `get_data.py`

External data is used for this discovery only. The scored leaderboard pipeline never reads it: its I/O guard refuses any non-permitted path, and that code is published separately once the leaderboard closes (31 October 2026).

Annex — Medicare nursing homes in wildfire country (secondary)

1,516 CMS-certified nursing homes lie in the scored tracts. 75 % have an Overture place of the same name within 250 m. The rate falls from 79.4 % in the least wildfire-exposed quarter of these homes (tract USFS WHP) to **68.3 %** in the most exposed one, where evacuating non-ambulatory residents is most time-critical. It does not fall with SVI (71 % in the least vulnerable quarter, 79 % in the most). Name matching undercounts renamed or chain facilities, so these are upper bounds on missing facilities (`discovery/cms_nursing_homes.py`, `out/cms_by_whp_quartile.csv`, `out/cms_by_svi_quartile.csv`).